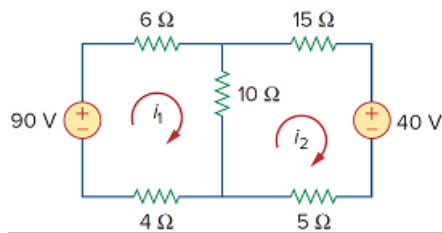


Problem

Calculate the mesh currents i_1 and i_2 of the circuit of Fig. 3.19.

Figure 3.19:



Step-by-step solution

Step 1 of 3

Refer to Figure 3.19 in the text book.

Apply mesh analysis in loop-1.

Applying Kirchhoff's voltage law, the algebraic sum of all voltages around a closed path (or loop) is zero.

$$-90 + 6i_1 + 10(i_1 - i_2) + 4i_1 = 0$$

$$-90 + 6i_1 + 10i_1 - 10i_2 + 4i_1 = 0$$

$$(6 + 10 + 4)i_1 - 10i_2 - 90 = 0$$

$$20i_1 - 10i_2 - 90 = 0$$

$$20i_1 = 10i_2 + 90$$

$$i_1 = \frac{10i_2 + 90}{20}$$

$$i_1 = 0.5i_2 + 4.5 \dots\dots (1)$$

Therefore, the expression for the current i_1 is $0.5i_2 + 4.5$.

Step 2 of 3

Apply mesh analysis in loop-2.

For mesh 2, Kirchhoff's voltage law gives,

$$40 + 15i_2 + 10(i_2 - i_1) + 5i_2 = 0$$

$$40 + 15i_2 + 10i_2 - 10i_1 + 5i_2 = 0$$

$$(15 + 10 + 5)i_2 - 10i_1 + 40 = 0$$

$$30i_2 - 10i_1 + 40 = 0$$

Substitute $0.5i_2 + 4.5$ for i_1 .

$$30i_2 - 10(0.5i_2 + 4.5) + 40 = 0$$

$$30i_2 - 5i_2 - 45 + 40 = 0$$

$$25i_2 - 5 = 0$$

$$25i_2 = 5$$

$$i_2 = \frac{5}{25}$$

$$= 200 \text{ mA}$$

Therefore, the value of the current i_2 is, 200 mA.

Recall equation (1).

$$i_1 = 0.5i_2 + 4.5$$

Substitute 200 mA for i_2 .

$$\begin{aligned} i_1 &= 0.5(200 \text{ m}) + 4.5 \\ &= 0.1 + 4.5 \\ &= 4.6 \text{ A} \end{aligned}$$

Therefore, the value of the current i_1 is, 4.6 A.